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Prop and Display Naming Conventions

These conventions ensure consistent naming across:

- **LOR sequencing (Channel Grid)**
- **Display database records (Postgres / SQLite)**
- **Physical labels used in the field**
- **Field wiring exports**

If naming rules are not followed, channels scatter in the grid, wiring reports drift from reality, and physical labels no longer match database records.

This document prevents chaos.

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1. Two Separate Naming Layers in LOR

LOR uses two different fields that serve different purposes:

Field	Purpose	Spaces Allowed	Used as Database Key
Name	Channel grid sorting & programming	Yes	No
Comment	Physical display identifier	No	Yes

These fields must not be confused.

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2. LOR Name Field (Channel Naming Convention)

The **Name field controls alphabetical sorting inside the LOR grid.**

LOR sorts strictly alphabetically.
Improper numbering or inconsistent formatting causes channels to scatter.

2.1 Required Structure

Recommended format:

```
<LL> <Group> <UID>-<CH> <Description>
```

Example:

```
TC 7B-01 Hippo Box
```

2.2 UID and Channel Rules (CRITICAL)

UID

- Controller ID in HEX
- Uppercase
- No leading zero padding
- Example: `7B`, `1F`, `20`

Channel

- Always 2-digit padded
- `01–16`
- Even if wiring exports show a single digit

Correct:

- `7B-01`
- `7B-02`
- `7B-10`

Incorrect:

- `7B-1`
- `7B-2`

If channels are not padded, alphabetical sorting breaks:

Wrong order:

- 7B-1
- 7B-10
- 7B-11
- 7B-2

Correct order:

- 7B-01
- 7B-02
- 7B-03
- 7B-10

2.3 Real Example (Hippo + Carolers)

Channel Grid Example

This grid shows:

- One multi-state animated display (Hippo)
- Three physical panels (CarolerPanel-01..03)
- Proper grouping by padded channel numbers

The naming works because:

- All channels begin with consistent stage prefix
- Channels are padded
- Related elements sort together

2.4 Unused Channels (SPARE Rule — REQUIRED)

Any unused channel on a controller must be explicitly added to the preview.

Do NOT leave gaps.

Unused channels must:

- Follow the exact same naming structure
- Use the same Stage and Group prefix
- Use correct HEX UID
- Use 2-digit padded channel number
- End with the word `Spare`

Example:

TC 7B-11 Spare
TC 7B-12 Spare

Why this is required:

- Makes unused capacity immediately visible in the Channel Grid
- Prevents accidental reuse of a channel already assumed to be empty
- Keeps controllers fully documented
- Allows quick identification of expansion space during programming

Unused channels must remain in the preview even if no physical wiring exists.

When a Spare channel is later assigned to a display:

- Rename the channel
- Update wiring documentation
- Remove the `Spare` designation

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3. Comment Field (Display Identifier)

The **Comment field defines the Display Name** used for:

- Physical labels
- Database joins
- Wiring exports
- Inventory tracking
- Maintenance records

This is the true identity of the physical display.

3.1 Comment Field Rules

- No spaces allowed
- Must start with 2-letter stage abbreviation
- Use hyphen as structural separator
- Must remain stable over time

Examples:

Standard stages:

- `FT-Arch-01`
- `RA-Arch-DS-01`
- `EC-Elf-P2-06`

GG stage (numeric prefix allowed):

- `GG20-Elden`
- `GG20-Elden-01`
- `GG30-V2Elden`
- `GG30-V2Elden-01`

3.2 Displays vs Sub-Props

Term	Definition
Display	One physical panel or unit
Sub-Prop	Logical motion element within a display

All sub-props must share the exact same **Comment value** as their parent display.

Example:

Channel Name	Comment
`TC 7B-01 Hippo Box`	`ChristmasHippo`
`TC 7B-04 Hippo Body Mid`	`ChristmasHippo`
`TC 7B-05 Hippo Body Full Head`	`ChristmasHippo`

3.3 Example – Multi-Panel Display (Caroler)

Visualization Example

Three physical panels:

- `CarolerPanel-01`
- `CarolerPanel-02`
- `CarolerPanel-03`

Each has its own Comment value.

Even if programming groups them visually, they remain separate physical units in inventory and wiring.

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4. Display Name Format (Comment Field)

`<Stage>-<DisplayName>-<Variation>-<Sequence>-<Color>`

Segment	Meaning
Stage	Two-letter stage code
DisplayName	CamelCase, no spaces
Variation	Optional location/version identifier
Sequence	Always 2-digit padded
Color	Optional suffix

Examples:

- `EC-Elf-P2-06`
- `RA-Arch-DS-01`
- `WF-MiniTree-G-04`
- `GG20-Elden-01`

4.1 DeviceType = Undetermined (Inventory-Only Displays)

Undetermined DeviceType Example

In the LOR preview editor, DeviceType is set to:

Undetermined

During SQLite parsing and Postgres ingestion, this becomes:

DeviceType = None

Operationally, both represent the same thing:

A physical inventory item with no controller assignment and no channel grid usage.

Use Cases

- Physical duplicates of programmed displays
- Arch supports
- Wrap stands
- Frame stands
- Future expansion inventory
- Volunteer Trailer Steps

- Igloo wagon for No Left Turn
- Attional displays were we duplicate the controllers for programming simplification like Emojis

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Quantity Creation (Max Circuits per Unit)

When creating inventory-only items:

- Set "Max Circuits per Unit" equal to the required quantity.
- LOR automatically generates suffixed records:
  - `-01`, `-02`, `-03`, etc.

Example:

Comment: `FC-WrapStand`

Max Circuits per Unit: `32`

Generated records:

- `FC-WrapStand-01`
- `FC-WrapStand-02`
- ...
- `FC-WrapStand-32`

If Max Circuits per Unit = 1:

- No suffix is added.

⚠ Critical Warning

The default value for "Max Circuits per Unit" is 16.

If not changed intentionally, LOR will create 16 inventory records automatically.

This will:

- Inflate inventory counts
- Pollute the database
- Require manual cleanup

Always verify this value before saving.

Rules

Inventory-only entries:

- Must follow full Comment naming rules
- Must not contain Network, UID, or ChannelGrid
- Must represent real physical inventory

These entries appear in database ingestion but do not affect wiring exports.

5. Field Wiring Alignment

Field Wiring Example

Field wiring exports rely on the Comment value.

If Comment naming is incorrect:

- Wiring sheets become inaccurate
- Displays cannot be located in the field
- Maintenance records break

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6. Inventory and Physical Labels

Every display label must:

- Match the Comment field exactly
- Be permanently attached to the display
- Match database and wiring documentation

The Comment value is the backbone of:

- Physical inventory
- Storage tracking
- Repair history
- Field setup instructions

☑ Summary

Field	Controls	Critical Rule
Name	Programming & Grid Sorting	Pad channels to 2 digits
Comment	Physical Identity	No spaces, stable, structured

Naming consistency prevents:

- Channel grid chaos
- Lost programming
- Wiring mismatches
- Inventory drift

> **Revision History**
>
> - GAL 25-10-29 — Initial merge
> - GAL 25-10-30 — Formatting cleanup

> - GAL 26-02-22 — Major clarification: separated Name vs Comment logic, added UID padding rule, added real-world examples with grid and wiring screenshots